

The Analytics Edge Data Competition 2026

Dates/Times

- Start of Kaggle competition: 24 July 2026 (12:00 pm Singapore time)
- End of Kaggle competition: 1 August 2026 (12:00 pm Singapore time)
- Submission of report: 10 August 2026 (12:00 pm Singapore time)

Kaggle Competition

Competition link: <https://www.kaggle.com/t/42a0b591922c450d99cb01ac7a662b66>.

Goal: In this competition, you will be predicting the choice among bundles of safety features in cars. Every observation has four bundles of safety features from which one bundle is selected. The data set has been discussed in class.

Data: The training data is provided in `train.csv` (total of 21565 observations) and the test data (total of 4997 observations) is provided in `test.csv`. The sample submission format is given in the file `sample_submission.csv`. You will need to submit your team's predictions in the format of this submission file.

Maximum daily submissions: Every team can make up to two submissions daily. Participants will need to wait until the next day after submitting the maximum number of daily submissions.

Evaluation: Evaluation of the predictions is done using the multi-class loss version of log-likelihood. In each observation, exactly one of the four alternatives (bundles) is chosen. For each observation, you must submit a predicted probability for each alternative. The metric that is computed to evaluate performance is the average negative log-likelihood of the model defined as follows:

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m y_{ij} \log(p_{ij})$$

where:

- n = Number of observations;
- m = Number of alternatives = 4;
- $\log$ = Natural logarithm;
- $y_{ij} = 1$ if for observation i , alternative j is chosen, and 0 otherwise;
- p_{ij} = Predicted probability that for observation i , alternative j is chosen.

The benchmark performance is to predict a probability of 1/4 for each alternative for each observation. In the leaderboard, this benchmark shows up as `benchmark.csv` with a score of:

$$\text{LogLoss of benchmark} = 1.38629$$

Lower the LogLoss metric of your model, the better is your performance. The values you submit need not necessarily sum up to 1 in the submission file. If it does not sum up to 1, the values will be automatically rescaled (each is divided by the sum) before evaluation. However this approach is not recommended since it may lead to numerical instabilities and it is suggested that you provide predicted probabilities that sum up to 1 for each observation.

Ranking: It is conceivable that teams could overfit a solution - which might be great for winning a competition, but not valuable for a real-world application. To prevent this, 70 percent of the observations in the test set will be used to evaluate performance on the public leaderboard and 30 percent of the observations in the test set will be used to evaluate performance on the private leaderboard. *This 70-30 split of the test set has been randomly created and is different year on year. So even if you have access to the best models that students in previous batches have created it does not mean that it will perform the best this year.*

You will be able to monitor your performance on the public leaderboard. However you will not know which observations are used for the public and private leaderboards respectively. The private leaderboard performance is kept a secret until after the competition deadline. Your final rankings are based on the private leaderboard. At the close of the competition, your best scoring submission on the public leaderboard (you do not need to select it) will be used to score the performance on the private leaderboard.

Team name: You are allowed only one Kaggle account per team. Name your team with the name of one team representative followed by the team's number. For example if the team representative is Walter White and team number is 1; name the team `walter_white_team_1`. You can reach out to the TA if you want to check on any of this.

You can only use this one account to submit your team's predictions. *If you use more than one account, it is unethical and reflect poorly on the values that you associate with education. I hope you value the learning process the most.* No private sharing of data outside your teams.

Codes: You can only use R for the competition (any R package is allowed). You are free to use methods that were not covered in the class. You are free to refer to online resources and use any tools (including AI assist). However you should only work on the projects within your groups and no help from other individuals is allowed.

Report

The report is a short document (maximum of 8 pages, any readable font size) which should contain: (i) a description of your best performing model on the public leaderboard (you can also add in a brief discussion on alternative models that you might have tried along the way), (ii) a discussion of your model fit both in the public and private leaderboard (if it is similar, worse or better), (iii) a discussion on insights you can gather from the model for this data and possible limitations of the approach. An executive summary is not needed. Please submit only one report per group. No appendix is allowed.

Grading

- 7 marks for performance on private leaderboard (as long as you beat the benchmark you will get at least 3 marks)
- 8 marks for performance on public leaderboard (as long as you beat the benchmark you will get at least 4 marks)
- 15 marks for report: (i) 5 marks; (ii) 2 marks; (iii) 5 marks, (iv) 3 marks for overall report quality